

D/3 'Panz matroy'

1.3.17.

$$A = \begin{pmatrix} 1 & -3 & 1 & -14 & 22 \\ -2 & 1 & 3 & 3 & -9 \\ -4 & -3 & 11 & -19 & 17 \end{pmatrix} \begin{matrix} \text{II} + 2\text{I} \\ \text{III} + 4\text{I} \end{matrix} \sim \begin{pmatrix} 1 & -3 & 1 & -14 & 22 \\ 0 & -5 & 5 & -25 & 35 \\ 0 & -15 & 15 & -75 & 105 \end{pmatrix} \begin{matrix} \\ \\ \text{III} - 3 \cdot \text{II} \end{matrix} \sim$$

$$\sim \begin{pmatrix} 1 & -3 & 1 & -14 & 22 \\ 0 & -5 & 5 & -25 & 35 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix} \Rightarrow r(A) = 2$$

1.3.18

$$A = \begin{pmatrix} 1 & 2 & 4 & -3 \\ 3 & 5 & 6 & -4 \\ 3 & 8 & 2 & -19 \end{pmatrix} \begin{matrix} \text{II} - 3\text{I} \\ \text{III} - 3\text{I} \end{matrix} \sim \begin{pmatrix} 1 & 2 & 4 & -3 \\ 0 & -1 & -6 & 5 \\ 0 & 2 & -10 & -10 \end{pmatrix} \begin{matrix} \\ \\ \text{III} + 2\text{II} \end{matrix} \sim$$

$$\sim \begin{pmatrix} 1 & 2 & 4 & -3 \\ 0 & -1 & -6 & 5 \\ 0 & 0 & -22 & 0 \end{pmatrix} \Rightarrow r(A) = 3$$

~~1.3.19~~

~~$$A = \begin{pmatrix} 3 & -1 & 3 & 2 & 5 \\ 5 & -3 & 2 & 3 & 4 \\ 1 & -3 & -5 & 0 & -7 \\ 7 & -5 & 1 & 4 & 1 \end{pmatrix} \begin{matrix} \text{I} - 3\text{III} \\ \text{II} - 3\text{I} \\ \text{IV} - \text{I} \end{matrix} \sim \begin{pmatrix} 3 & -1 & 3 & 2 & 5 \\ 5 & -3 & 2 & 3 & 4 \\ 1 & -3 & -5 & 0 & -7 \\ 2 & -2 & -1 & 1 & -3 \end{pmatrix} \begin{matrix} \text{II} - 2\text{IV} \\ \\ \\ \end{matrix} \sim$$~~
~~$$\sim \begin{pmatrix} 0 & 8 & 18 & 2 & 26 \\ 1 & 3 & 12 & 3 & 18 \\ 1 & -3 & -5 & 0 & -7 \\ 2 & -2 & -1 & 1 & -3 \end{pmatrix} \begin{matrix} \text{II} - \text{III} \\ \text{IV} - 2\text{III} \\ \\ \end{matrix} \sim \begin{pmatrix} 0 & 8 & 18 & 2 & 26 \\ 0 & 6 & 17 & 3 & 25 \\ 1 & -3 & -5 & 0 & -7 \\ 0 & 4 & 9 & 1 & 11 \end{pmatrix} \begin{matrix} \text{I} \cdot \frac{1}{2} \\ \text{II} - \text{IV} \\ \\ \end{matrix} \sim$$~~
~~$$\sim \begin{pmatrix} 0 & 4 & 9 & 1 & 11 \\ 0 & 2 & 8 & 2 & 14 \\ 1 & -3 & -5 & 0 & -7 \\ 0 & 4 & 9 & 1 & 11 \end{pmatrix} \begin{matrix} \text{I} - \text{IV} \\ \text{II} - \text{IV} \cdot \frac{1}{2} \\ \\ \end{matrix} \sim \begin{pmatrix} 0 & 0 & 0 & 0 & 2 \\ 0 & 0 & 3,5 & 1,5 & 8,5 \\ 1 & -3 & -5 & 0 & -7 \\ 0 & 4 & 9 & 1 & 11 \end{pmatrix} \begin{matrix} \uparrow \downarrow \\ \uparrow \downarrow \\ \\ \end{matrix} \sim$$~~

$$\begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

1 3 1 9

$$A = \begin{pmatrix} 3 & -1 & 3 & 2 & 5 \\ 5 & -3 & 2 & 3 & 4 \\ 1 & -3 & -5 & 0 & -7 \\ 7 & -5 & 1 & 4 & 1 \end{pmatrix} \sim \begin{pmatrix} 1 & -1/3 & 1 & 2/3 & 5/3 \\ 5 & -3 & 2 & 3 & 4 \\ 1 & -3 & -5 & 0 & -7 \\ 7 & -5 & 1 & 4 & 1 \end{pmatrix} \begin{matrix} I \cdot \frac{1}{3} \\ II - 5I \\ III - I \\ IV - 7I \end{matrix}$$

$$\sim \begin{pmatrix} 1 & -1/3 & 1 & 2/3 & 5/3 \\ 0 & -4/3 & -3 & -1/3 & -13/3 \\ 0 & -8/3 & -6 & -2/3 & -26/3 \\ 0 & -8/3 & -6 & -2/3 & -32/3 \end{pmatrix} \begin{matrix} \\ III - 2II \\ IV - 2II \end{matrix} \sim \begin{pmatrix} 1 & -1/3 & 1 & 2/3 & 5/3 \\ 0 & -4/3 & -3 & -1/3 & -13/3 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -2 \end{pmatrix} \begin{matrix} \\ \\ V \cdot 2 \end{matrix}$$

$$\sim \begin{pmatrix} 1 & -1/3 & 1 & 2/3 & 5/3 \\ 0 & -4/3 & -3 & -1/3 & -13/3 \\ 0 & 0 & 0 & 0 & -2 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix} \Rightarrow r(A) = 3$$

1 3 20

$$A = \begin{pmatrix} 24 & 19 & 36 & 72 & -38 \\ 49 & 40 & 73 & 142 & -80 \\ 73 & 59 & 98 & 219 & -118 \\ 47 & 36 & 71 & 141 & -72 \end{pmatrix} \sim \begin{matrix} I \cdot 1/24 \\ \\ \\ \end{matrix}$$

$$\sim \begin{pmatrix} 1 & 19/24 & 36/24 & 72/24 & -38/24 \\ 49 & 40 & 73 & 142 & -80 \\ 73 & 59 & 98 & 219 & -118 \\ 47 & 36 & 71 & 141 & -72 \end{pmatrix} \begin{matrix} \\ II - 49I \\ III - 73I \\ IV - 47I \end{matrix}$$

$$\sim \begin{pmatrix} 1 & 19/24 & 36/24 & 72/24 & -38/24 \\ 0 & 29/24 & -1/2 & 0 & -29/12 \\ 0 & 29/24 & -23/2 & 0 & -29/12 \\ 0 & -29/24 & 1/2 & 0 & 29/12 \end{pmatrix} \begin{matrix} \\ \\ II \cdot 24/29 \\ \\ \end{matrix}$$

$$\sim \begin{pmatrix} 1 & 19/24 & 36/24 & 72/24 & -38/24 \\ 0 & 1 & -12/24 & 0 & -2 \\ 0 & 29/24 & -23/2 & 0 & -29/12 \\ 0 & -29/24 & 1/2 & 0 & 29/12 \end{pmatrix} \begin{array}{l} \\ \\ \text{III} - (29/24) \cdot \text{II} \\ \text{IV} + (29/24) \cdot \text{II} \end{array} \sim$$

$$\sim \begin{pmatrix} 1 & 19/24 & 36/24 & 72/24 & -38/24 \\ 0 & 1 & -12/24 & 0 & -2 \\ 0 & 0 & -11 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix} \Rightarrow r(A) = 3$$

1.3.21.

$$A = \begin{pmatrix} 4 & 3 & -5 & 2 & 3 \\ 8 & 6 & -7 & 4 & 2 \\ 4 & 3 & -8 & 2 & 7 \\ 4 & 3 & 1 & 2 & -5 \\ 8 & 6 & -1 & 4 & -6 \end{pmatrix} \begin{array}{l} \text{I} \cdot 1/4 \\ \\ \\ \\ \end{array} \sim \begin{pmatrix} 1 & 3/4 & -5/4 & 1/2 & 3/4 \\ 8 & 6 & -7 & 4 & 2 \\ 4 & 3 & -8 & 2 & 7 \\ 4 & 3 & 1 & 2 & -5 \\ 8 & 6 & -1 & 4 & -6 \end{pmatrix} \begin{array}{l} \\ \text{II} - 8 \cdot \text{I} \\ \text{III} - 4 \cdot \text{I} \\ \text{IV} - 4 \cdot \text{I} \\ \text{V} - 8 \cdot \text{I} \end{array} \sim$$

$$\sim \begin{pmatrix} 1 & 3/4 & -5/4 & 1/2 & 3/4 \\ 0 & 0 & 3 & 0 & -4 \\ 0 & 0 & 13 & 0 & 4 \\ 0 & 0 & 6 & 0 & -8 \\ 0 & 0 & 9 & 0 & -12 \end{pmatrix} \begin{array}{l} \\ \text{II} \cdot 1/3 \\ \\ \\ \end{array} \sim \begin{pmatrix} 1 & 3/4 & -5/4 & 1/2 & 3/4 \\ 0 & 0 & 1 & 0 & -4/3 \\ 0 & 0 & 13 & 0 & 4 \\ 0 & 0 & 6 & 0 & -8 \\ 0 & 0 & 9 & 0 & -12 \end{pmatrix} \begin{array}{l} \\ \\ \text{III} - 13 \cdot \text{II} \\ \text{IV} - 6 \cdot \text{II} \\ \text{V} - 9 \cdot \text{II} \end{array} \sim$$

$$\sim \begin{pmatrix} 1 & 3/4 & -5/4 & 1/2 & 3/4 \\ 0 & 0 & 1 & 0 & -4/3 \\ 0 & 0 & 0 & 0 & 64/3 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix} \Rightarrow r(A) = 3$$

1.5.22

$$A = \begin{pmatrix} 17 & -28 & 45 & 11 & 39 \\ 24 & -37 & 61 & 13 & 50 \\ 25 & -7 & 32 & -18 & -11 \\ 31 & 12 & 19 & -43 & -55 \\ 42 & 13 & 29 & -55 & -68 \end{pmatrix} \begin{array}{l} \text{I} \cdot 1/17 \\ \\ \\ \\ \end{array} \sim$$

$$\sim \begin{pmatrix} 1 & -1^{1/17} & 2^{1/17} & 1^{1/17} & 2^{5/17} \\ 24 & -37 & 61 & 13 & 50 \\ 25 & -7 & 52 & -18 & -4 \\ 31 & 12 & 19 & -43 & -55 \\ 42 & 13 & 29 & -55 & -68 \end{pmatrix} \begin{array}{l} \text{II} - 24\text{I} \\ \text{III} - 25\text{I} \\ \text{IV} - 31\text{I} \\ \text{V} - 42\text{I} \end{array} \sim$$

$$\sim \begin{pmatrix} 1 & -1^{1/17} & 2^{1/17} & 1^{1/17} & 2^{5/17} \\ 0 & 2^{9/17} & -2^{9/17} & -2^{9/17} & -5^{1/17} \\ 0 & 34^{3/17} & -34^{3/17} & -34^{3/17} & -68^{6/17} \\ 0 & 63^{1/17} & -63^{1/17} & -63^{1/17} & -126^{2/17} \\ 0 & 82^{3/17} & -82^{3/17} & -82^{3/17} & -164^{6/17} \end{pmatrix} \begin{array}{l} \text{II} = 2^{9/17} \\ \\ \\ \\ \end{array} \sim$$

$$\sim \begin{pmatrix} 1 & -1^{1/17} & 2^{1/17} & 1^{1/17} & 2^{5/17} \\ 0 & 1 & -1 & -1 & -2 \\ 0 & 34^{3/17} & -34^{3/17} & -34^{3/17} & -68^{6/17} \\ 0 & 63^{1/17} & -63^{1/17} & -63^{1/17} & -126^{2/17} \\ 0 & 82^{3/17} & -82^{3/17} & -82^{3/17} & -164^{6/17} \end{pmatrix} \begin{array}{l} \\ \text{III} - 34^{3/17} \\ \text{IV} - 63^{1/17} \\ \text{V} - 82^{3/17} \end{array} \sim$$

$$\sim \begin{pmatrix} 1 & -1^{1/17} & 2^{1/17} & 1^{1/17} & 2^{5/17} \\ 0 & 1 & -1 & -1 & -2 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix} \Rightarrow r(A) = 2$$

1. 3. 2. 3

$$A = \begin{pmatrix} 5 & -1 & 2 \\ 4 & -3 & 3 \\ 1 & 3 & 0 \end{pmatrix}$$

1) $M_1^{(1)} = |a_{11}| = |5| = 5 \neq 0 \Rightarrow r(A) \geq 1$

2) $M_2^{(1)} = \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} = \begin{vmatrix} 5 & -1 \\ 4 & -3 \end{vmatrix} = -9 + 4 = -5 \neq 0 \Rightarrow r(A) \geq 2$

3) $M_3^{(1)} = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = \begin{vmatrix} 5 & -1 & 2 \\ 4 & -3 & 3 \\ 1 & 3 & 0 \end{vmatrix} = \begin{vmatrix} + & - & + \\ - & + & - \\ + & - & + \end{vmatrix} = 1 \cdot \begin{vmatrix} -1 & 2 \\ 3 & 3 \end{vmatrix} - 3 \cdot \begin{vmatrix} 5 & 2 \\ 4 & 3 \end{vmatrix} + 0 \cdot \begin{vmatrix} 5 & -1 \\ 4 & -3 \end{vmatrix} =$

$$= -3 + 6 - 3(9 - 8) + 0 = 0 = 0 \Rightarrow r(A) < 3$$

$$4) \left. \begin{array}{l} r(A) \geq 2 \\ r(A) < 3 \end{array} \right\} \Rightarrow \underline{r(A) = 2}$$

5) Баунсний минор:

$$M_2^{(1)} = \begin{vmatrix} 3 & -1 \\ 4 & -3 \end{vmatrix}$$

1. 3. 2. 4

$$A = \begin{pmatrix} 3 & -1 & 2 \\ 4 & -3 & 3 \\ 1 & 3 & 2 \end{pmatrix}$$

$$1) M_1^{(1)} = |a_{11}| = 3 \neq 0 \Rightarrow r(A) \geq 1$$

$$2) M_2^{(1)} = \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} = \begin{vmatrix} 3 & -1 \\ 4 & -3 \end{vmatrix} = -9 + 4 = -5 \neq 0 \Rightarrow r(A) \geq 2$$

$$3) M_3^{(1)} = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = \begin{vmatrix} 3 & -1 & 2 \\ 4 & -3 & 3 \\ 1 & 3 & 2 \end{vmatrix} = \begin{vmatrix} + & - & + \\ - & + & - \\ + & - & + \end{vmatrix} = 1 \cdot \begin{vmatrix} -1 & 2 \\ -3 & 3 \end{vmatrix} - 3 \cdot \begin{vmatrix} 3 & 2 \\ 4 & 3 \end{vmatrix} + 2 \cdot \begin{vmatrix} 3 & -1 \\ 4 & -3 \end{vmatrix} =$$

$$= -3 + 6 - 3(9 - 8) + 2(-9 + 4) = 3 - 3 - 10 = -10 \neq 0 \Rightarrow \underline{r(A) = 3}$$

4) Баунсний минор:

$$M_3^{(1)} = \begin{vmatrix} 3 & -1 & 2 \\ 4 & -3 & 3 \\ 1 & 3 & 2 \end{vmatrix}$$

1. 3. 2. 5

$$A = \begin{pmatrix} 2 & -1 & 5 & 6 \\ 1 & 1 & 3 & 5 \\ 1 & -5 & 1 & -3 \end{pmatrix}$$

$$1) M_1^{(1)} = |a_{11}| = |2| = 2 \neq 0 \Rightarrow r(A) \geq 1$$

$$2) M_2^{(1)} = \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} = \begin{vmatrix} 2 & -1 \\ 1 & 1 \end{vmatrix} = 2 + 1 = 3 \neq 0 \Rightarrow r(A) \geq 2$$

$$3) M_3^{(1)} = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = \begin{vmatrix} 2 & -1 & 5 \\ 1 & 1 & 3 \\ 1 & -5 & 1 \end{vmatrix} = \begin{vmatrix} + & - & + \\ - & + & - \\ + & - & + \end{vmatrix} = 1 \cdot \begin{vmatrix} -1 & 5 \\ 1 & 3 \end{vmatrix} + 5 \cdot \begin{vmatrix} 2 & 5 \\ 1 & 3 \end{vmatrix} + 1 \cdot \begin{vmatrix} 2 & -1 \\ 1 & 1 \end{vmatrix} =$$

$$= -3 - 5 + 5(6 - 5) + 2 + 1 = -8 + 5 + 3 = 0$$

$$M_5^{(1)} = \begin{vmatrix} a_{11} & a_{12} & a_{14} \\ a_{21} & a_{22} & a_{24} \\ a_{31} & a_{32} & a_{34} \end{vmatrix} = \begin{vmatrix} 2 & -1 & 6 \\ 1 & 1 & 5 \\ 1 & -5 & -3 \end{vmatrix} = \begin{vmatrix} + & - & + \\ - & + & - \\ + & - & + \end{vmatrix} = 1 \cdot \begin{vmatrix} -1 & 6 \\ 1 & 5 \end{vmatrix} + 5 \cdot \begin{vmatrix} 2 & 6 \\ 1 & 5 \end{vmatrix} - 3 \cdot \begin{vmatrix} 2 & -1 \\ 1 & 1 \end{vmatrix} =$$

$$= -5 - 6 + 5(10 - 6) - 3(2 + 1) = -11 + 20 - 9 = 0 \Rightarrow r(A) < 3$$

$$4) \begin{cases} r(A) < 3 \\ r(A) \geq 2 \end{cases} \Rightarrow r(A) = 2$$

5) Базисный минор:

$$M_2^{(1)} = \begin{vmatrix} 2 & -1 \\ 1 & 1 \end{vmatrix}$$

1.3.2.6

$$A = \begin{pmatrix} 1 & -2 & 3 & -4 & 4 \\ 0 & 1 & -1 & 1 & -3 \\ 1 & 3 & 0 & -3 & 1 \\ 0 & -7 & 3 & 1 & -3 \end{pmatrix}$$

$$1) M_1^{(1)} = |a_{11}| = |1| = 1 \neq 0 \Rightarrow r(A) \geq 1$$

$$2) M_2^{(1)} = \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} = \begin{vmatrix} 1 & -2 \\ 0 & 1 \end{vmatrix} = 1 - 0 = 1 \neq 0 \Rightarrow r(A) \geq 2$$

$$3) M_3^{(1)} = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = \begin{vmatrix} 1 & -2 & 3 \\ 0 & 1 & -1 \\ 1 & 3 & 0 \end{vmatrix} = \begin{vmatrix} + & - & + \\ - & + & - \\ + & - & + \end{vmatrix} = 0 \cdot \begin{vmatrix} -2 & 3 \\ 3 & 0 \end{vmatrix} + 1 \cdot \begin{vmatrix} 1 & 3 \\ 1 & 0 \end{vmatrix} + 1 \cdot \begin{vmatrix} 1 & -2 \\ 1 & 3 \end{vmatrix} =$$

$$= 0 - 9 + 5 = -4 \neq 0 \Rightarrow r(A) \geq 3$$

$$4) M_4^{(1)} = \begin{pmatrix} 1 & -2 & 3 & -4 \\ 0 & 1 & -1 & 1 \\ 1 & 3 & 0 & -3 \\ 0 & -7 & 3 & 1 \end{pmatrix} = \begin{pmatrix} + & - & + & - \\ - & + & - & + \\ + & - & + & - \\ - & + & - & + \end{pmatrix} = 1 \cdot \begin{pmatrix} 1 & -1 & 1 \\ 3 & 0 & -3 \\ -7 & 3 & 1 \end{pmatrix} - 0 \cdot \begin{pmatrix} -2 & 3 & -4 \\ 3 & 0 & -3 \\ -7 & 3 & 1 \end{pmatrix} +$$

$$+ 1 \cdot \begin{pmatrix} -2 & 3 & -4 \\ 1 & -1 & 1 \\ -7 & 3 & 1 \end{pmatrix} - 0 \cdot \begin{pmatrix} -2 & 3 & -4 \\ 1 & -1 & 1 \\ 3 & 0 & -3 \end{pmatrix} = (1 \cdot 0 \cdot 1 + 3 \cdot 3 \cdot 1 + (-1) \cdot (-3) \cdot (-7) - 1 \cdot 0 \cdot (-7) -$$

$$- 3 \cdot (-3) \cdot 1 - 3 \cdot (-1) \cdot 1) - 0 + ((-2) \cdot (-1) \cdot 1 +$$

$$+ 1 \cdot 3 \cdot (-4) + 3 \cdot 1 \cdot (-7) - (-4) \cdot (-1) \cdot (-7) - 3 \cdot 1 \cdot (-2) - 1 \cdot 3 \cdot 1) - 0 =$$

$$= (0 + 9 - 21 - 0 + 9 + 3) + (2 - 12 - 21 + 28 + 6 - 3) =$$

$$= 0 + 0 = 0$$

$$M_4^{(1)} = \begin{pmatrix} 1 & -2 & 3 & 4 \\ 0 & 1 & -1 & -3 \\ 1 & 3 & 0 & 1 \\ 0 & -7 & 3 & -3 \end{pmatrix} = \begin{pmatrix} + & - & + & - \\ - & + & - & + \\ + & - & + & - \\ - & + & - & + \end{pmatrix} = 1 \cdot \begin{pmatrix} 1 & -1 & -3 \\ 3 & 0 & 1 \\ -7 & 3 & -3 \end{pmatrix} - 0 \cdot \begin{pmatrix} -2 & 3 & 4 \\ 3 & 0 & 1 \\ -7 & 3 & -3 \end{pmatrix} +$$

$$+ 1 \cdot \begin{pmatrix} -2 & 3 & 4 \\ 1 & -1 & -3 \\ -7 & 3 & -3 \end{pmatrix} - 0 \cdot \begin{pmatrix} -2 & 3 & 4 \\ 1 & -1 & -3 \\ 3 & 0 & 1 \end{pmatrix} = (1 \cdot 0 \cdot (-3) + (-1) \cdot 1 \cdot (-7) + 3 \cdot 3 \cdot (-3) - (-3) \cdot 0 \cdot (-7) -$$

$$- 3 \cdot 1 \cdot 1 - 3 \cdot (-1) \cdot (-3)) - 0 + ((-2) \cdot (-1) \cdot (-3) + 1 \cdot 3 \cdot 4 + 3 \cdot (-3) \cdot (-7) - 4 \cdot (-1) \cdot (-7) -$$

$$- 3 \cdot (-3) \cdot (-2) - 1 \cdot 3 \cdot (-3)) - 0 = (0 + 7 - 27 - 0 - 3 - 9) + (-6 + 12 + 63 -$$

$$- 28 - 18 + 9) = -32 + 32 = 0 \Rightarrow r(A) < 4$$

$$5) \begin{matrix} r(A) < 4 \\ r(A) \geq 3 \end{matrix} \Rightarrow r(A) = 3$$

6) Базисный минор:

$$M_3^{(1)} = \begin{vmatrix} 1 & -2 & 3 \\ 0 & 1 & -1 \\ 1 & 3 & 0 \end{vmatrix}$$

1. 3. 2. 7.

$$A = \begin{pmatrix} 1 & -2 & 1 & -1 & 1 \\ 2 & 1 & -1 & 2 & -3 \\ 3 & -2 & -1 & 1 & -2 \\ 2 & -5 & 1 & -2 & 2 \end{pmatrix}$$

$$1) M_1^{(1)} = |a_{11}| = |1| = 1 \neq 0 \Rightarrow r(A) \geq 1$$

$$2) M_2^{(1)} = \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} = \begin{vmatrix} 1 & -2 \\ 2 & 1 \end{vmatrix} = 1 + 4 = 5 \neq 0 \Rightarrow r(A) \geq 2$$

$$3) M_3^{(1)} = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = \begin{vmatrix} 1 & -2 & 1 \\ 2 & 1 & -1 \\ 3 & -2 & -1 \end{vmatrix} = 1 \cdot 1 \cdot (-1) + 2 \cdot (-2) \cdot 1 + (-2) \cdot (-1) \cdot 3 -$$

$$- 1 \cdot 1 \cdot 3 - (-2) \cdot (-1) \cdot 1 - 2 \cdot (-2) \cdot (-1) = -1 - 4 + 6 - 3 - 2 - 4 = -8 \neq 0 \Rightarrow$$

$$\Rightarrow r(A) \geq 3$$

$$4) M_4^{(1)} = \begin{vmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \\ a_{41} & a_{42} & a_{43} & a_{44} \end{vmatrix} = \begin{vmatrix} 1 & -2 & 1 & -1 \\ 2 & 1 & -1 & 2 \\ 3 & -2 & -1 & 1 \\ 2 & -5 & 1 & -2 \end{vmatrix} = \begin{vmatrix} + & - & + & - \\ - & + & - & + \\ + & - & + & - \\ - & + & - & + \end{vmatrix}$$

$$= 1 \cdot \begin{vmatrix} 2 & 1 & -1 \\ 3 & -2 & -1 \\ 2 & -5 & 1 \end{vmatrix} + 2 \cdot \begin{vmatrix} 1 & -2 & 1 \\ 3 & -2 & -1 \\ 2 & -5 & 1 \end{vmatrix} - 1 \cdot \begin{vmatrix} 1 & -2 & 1 \\ 2 & 1 & -1 \\ 2 & -5 & 1 \end{vmatrix} - 2 \cdot \begin{vmatrix} 1 & -2 & 1 \\ 2 & 1 & -1 \\ 3 & -2 & -1 \end{vmatrix} =$$

$$= (2 \cdot (-2) \cdot 1 + 3 \cdot (-5) \cdot (-1) + 1 \cdot (-1) \cdot 2 - (-1) \cdot (-2) \cdot 2 - 1 \cdot 3 \cdot 1 - (-1) \cdot (-5) \cdot 2) +$$

$$+ 2 \cdot (1 \cdot (-2) \cdot 1 + 3 \cdot (-5) \cdot 1 + (-2) \cdot (-1) \cdot 2 - 1 \cdot (-2) \cdot 2 - (-2) \cdot 3 \cdot 1 - (-1) \cdot (-5) \cdot 1) -$$

$$- (1 \cdot 1 \cdot 1 + (-2) \cdot (-1) \cdot 2 + 2 \cdot (-5) \cdot 1 - 1 \cdot 1 \cdot 2 - 2 \cdot (-2) \cdot 1 - (-1) \cdot (-5) \cdot 1) -$$

$$- 2 \cdot (1 \cdot 1 \cdot (-1) + 2 \cdot (-2) \cdot 1 + (-2) \cdot (-1) \cdot 3 - 1 \cdot 1 \cdot 3 - 2 \cdot (-2) \cdot (-1) - (-2) \cdot (-1) \cdot 1) =$$

$$= -8 + 2 \cdot (-8) - (-8) - 2 \cdot (-8) = 0$$

~~2. 4. 3. 0~~

$$M_4^{(2)} = \begin{vmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \\ a_{41} & a_{42} & a_{43} & a_{44} \end{vmatrix} = \begin{vmatrix} 1 & -2 & 1 & 1 \\ 2 & 1 & -1 & -3 \\ 3 & -2 & -1 & -2 \\ 2 & -5 & 1 & 2 \end{vmatrix} = \begin{vmatrix} + & - & + & - \\ - & + & - & + \\ + & - & + & - \\ - & + & - & + \end{vmatrix} =$$

$$= -1 \cdot \begin{vmatrix} 2 & 1 & -1 \\ 3 & -2 & -1 \\ 2 & -5 & 1 \end{vmatrix} - 3 \cdot \begin{vmatrix} 1 & -2 & 1 \\ 3 & -2 & -1 \\ 2 & -5 & 1 \end{vmatrix} + 2 \cdot \begin{vmatrix} 1 & -2 & 1 \\ 2 & 1 & -1 \\ 2 & -5 & 1 \end{vmatrix} + \begin{vmatrix} 1 & -2 & 1 \\ 2 & 1 & -1 \\ 3 & -2 & -1 \end{vmatrix} =$$

$$= -(2 \cdot (-2) \cdot 1 + (-1) \cdot 3 \cdot (-5) + 2 \cdot 1 \cdot (-1) - (-1) \cdot (-2) \cdot 2 - 3 \cdot 1 \cdot 1 - 2 \cdot (-1) \cdot (-5)) -$$

$$- 3(1 \cdot (-2) \cdot 1 + 3 \cdot (-5) \cdot 1 + (-2) \cdot (-1) \cdot 2 - 1 \cdot (-2) \cdot 2 - (-2) \cdot 3 \cdot 1 - (-1) \cdot (-5) \cdot 1) +$$

$$+ 2(1 \cdot 1 \cdot 1 + 2 \cdot (-2) \cdot (-1) + 1 \cdot 2 \cdot (-5) - 1 \cdot 2 \cdot (-1) \cdot (-5) - 1 \cdot 2 \cdot (-2)) +$$

$$+ 2(1 \cdot 1 \cdot (-1) + 2 \cdot (-2) \cdot 1 + (-2) \cdot (-1) \cdot 3 - 1 \cdot 3 \cdot 2 - (-2) \cdot (-1) \cdot (-2) - (-1) \cdot 1 \cdot 1) =$$

$$= -(-8) - 3 \cdot (-8) + 2 \cdot (-8) + 2 \cdot (-8) = 0 \Rightarrow r(A) \leq 4$$

$$\begin{matrix} 5) & r(A) \geq 3 \\ & r(A) < 4 \end{matrix} \Rightarrow r(A) = 3$$

6) Бажисный минор:

$$M_3^{(1)} = \begin{vmatrix} 1 & -2 & 1 \\ 2 & 1 & -1 \\ 3 & -2 & -1 \end{vmatrix}$$

1. 3. 28

$$A = \begin{pmatrix} 2 & 1 & -1 & -1 & 1 \\ 1 & -1 & 1 & 1 & -2 \\ 3 & 3 & -3 & -3 & 4 \\ 4 & 5 & -5 & -5 & 7 \end{pmatrix}$$

$$1) M_1^{(1)} = |a_{11}| = |2| = 2 \neq 0 \Rightarrow r(A) \geq 1$$

$$2) M_2^{(1)} = \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} = \begin{vmatrix} 2 & 1 \\ 1 & -1 \end{vmatrix} = -2 \cdot 1 - 1 \cdot 1 = -3 \neq 0 \Rightarrow r(A) \geq 2$$

$$3) M_3^{(1)} = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = \begin{vmatrix} 2 & 1 & -1 \\ 1 & -1 & 1 \\ 3 & 3 & -3 \end{vmatrix} = 2 \cdot (-1) \cdot (-3) + 1 \cdot 1 \cdot 3 + 1 \cdot 3 \cdot (-1) - (-1) \cdot (-1) \cdot 3 - 1 \cdot 1 \cdot (-3) - 1 \cdot 3 \cdot 2 = 0$$

$$M_3^{(2)} = \begin{vmatrix} a_{11} & a_{12} & a_{14} \\ a_{21} & a_{22} & a_{24} \\ a_{31} & a_{32} & a_{34} \end{vmatrix} = \begin{vmatrix} 2 & 1 & -1 \\ 1 & -1 & 1 \\ 3 & 3 & -3 \end{vmatrix} = 2(-1)(-3) + 1(-3) + 3(-1) - (-1)(-1) \cdot 3 - 1(-1)(-3) - 1 \cdot 3 \cdot 2 = 0$$

$$M_3^{(5)} = \begin{vmatrix} a_{11} & a_{12} & a_{15} \\ a_{21} & a_{22} & a_{25} \\ a_{31} & a_{32} & a_{35} \end{vmatrix} = \begin{vmatrix} 2 & 1 & 1 \\ 1 & -1 & -2 \\ 3 & 3 & 4 \end{vmatrix} = 2(-1) \cdot 4 + 1 \cdot 3 \cdot 1 + 1 \cdot (-2) \cdot 3 - 1(-1) \cdot 3 - 1 \cdot 1 \cdot 4 - (-2) \cdot 3 \cdot 2 = 0$$

$$M_3^{(4)} = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{41} & a_{42} & a_{43} \end{vmatrix} = \begin{vmatrix} 2 & 1 & -1 \\ 1 & -1 & 1 \\ 4 & 5 & -5 \end{vmatrix} = 2(-1)(-5) + 1 \cdot 1 \cdot 4 + 1 \cdot 5 \cdot (-1) - (-1)(-1) \cdot 4 - 1 \cdot 1 \cdot (-5) - 1 \cdot 5 \cdot 2 = 0$$

$$M_3^{(5)} = \begin{vmatrix} a_{11} & a_{12} & a_{14} \\ a_{21} & a_{22} & a_{24} \\ a_{41} & a_{42} & a_{44} \end{vmatrix} = \begin{vmatrix} 2 & 1 & -1 \\ 1 & -1 & 1 \\ 4 & 5 & -5 \end{vmatrix} = M_3^{(4)} = 0$$

$$M_3^{(6)} = \begin{vmatrix} a_{11} & a_{12} & a_{15} \\ a_{21} & a_{22} & a_{25} \\ a_{41} & a_{42} & a_{45} \end{vmatrix} = \begin{vmatrix} 2 & 1 & 1 \\ 1 & -1 & -2 \\ 4 & 5 & 7 \end{vmatrix} = 2(-1) \cdot 7 + 1 \cdot 5 \cdot 1 + 1 \cdot (-2) \cdot 4 - 1(-1) \cdot 4 - 1 \cdot 1 \cdot 7 - 2 \cdot 5 \cdot (-2) = 0 \Rightarrow r(A) < 3$$

$$\Rightarrow \begin{matrix} r(A) \geq 2 \\ r(A) < 3 \end{matrix} \Rightarrow r(A) = 2$$

4) Баисный минор:

$$M_2^{(1)} = \begin{vmatrix} 2 & 1 \\ 1 & -1 \end{vmatrix}$$

1.3.29

$$A = \begin{pmatrix} 1 & -3 & 2 & 0 \\ 2 & -3 & -1 & 3 \\ 3 & -6 & -1 & \lambda \\ 1 & -2 & 0 & 1 \end{pmatrix} \begin{matrix} \\ \text{II} - 2\text{I} \\ \text{III} - 3\text{I} \\ \text{IV} - \text{I} \end{matrix} \sim \begin{pmatrix} 1 & -3 & 2 & 0 \\ 0 & 3 & -5 & 3 \\ 0 & 3 & -7 & \lambda \\ 0 & 1 & -2 & 1 \end{pmatrix} \begin{matrix} \\ \\ \text{III} - \text{II} \\ 3\text{IV} - \text{II} \end{matrix} \sim$$

$$\sim \begin{pmatrix} 1 & -3 & 2 & 0 \\ 0 & 3 & -5 & 3 \\ 0 & 0 & -2 & \lambda - 3 \\ 0 & 0 & -1 & 0 \end{pmatrix} \sim \begin{pmatrix} 1 & -3 & 2 & 0 \\ 0 & 3 & -5 & 3 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & -2 & \lambda - 3 \end{pmatrix} \begin{matrix} \\ \\ \\ \text{IV} - 2 \cdot \text{III} \end{matrix} \sim$$

$$\sim \begin{pmatrix} 1 & -3 & 2 & 0 \\ 0 & 3 & -5 & 3 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & \lambda-3 \end{pmatrix}$$

При $\lambda-3=0 \Rightarrow \lambda=3 \Rightarrow$ IV строка станет нулевой строкой $\Rightarrow$
 $\Rightarrow r(A)=3$

Ответ: при $\lambda=3$ $r(A)=3$
 при $\lambda \neq 3$ $r(A)=4$

1.3.30

$$A = \begin{pmatrix} 3 & 1 & 1 & 4 \\ \lambda & 4 & 10 & 1 \\ 1 & 7 & 17 & 3 \\ 2 & 2 & 4 & 3 \end{pmatrix} \sim \begin{pmatrix} 1 & 7 & 17 & 3 \\ 2 & 2 & 4 & 3 \\ 3 & 1 & 1 & 4 \\ \lambda & 4 & 10 & 1 \end{pmatrix} \begin{matrix} \\ \text{II}-2\text{I} \\ \text{III}-3\text{I} \\ \text{IV}-\lambda\text{I} \end{matrix} \sim$$

$$\sim \begin{pmatrix} 1 & 7 & 17 & 3 \\ 0 & -12 & -30 & -3 \\ 0 & -20 & -50 & -5 \\ 0 & 4-7\lambda & 10-17\lambda & 1-3\lambda \end{pmatrix} \begin{matrix} \\ \\ \text{III} \cdot \frac{3}{5} - \text{II} \\ \end{matrix} \sim \begin{pmatrix} 1 & 7 & 17 & 3 \\ 0 & -12 & -30 & -3 \\ 0 & 0 & 0 & 0 \\ 0 & 4-7\lambda & 10-17\lambda & 1-3\lambda \end{pmatrix} \sim$$

$$\sim \begin{pmatrix} 1 & 7 & 17 & 3 \\ 0 & -12 & -30 & -3 \\ 0 & 4-7\lambda & 10-17\lambda & 1-3\lambda \\ 0 & 0 & 0 & 0 \end{pmatrix} \begin{matrix} \\ \\ 12 \cdot \text{III} + \text{II} \cdot (4-7\lambda) \\ \end{matrix} \sim \begin{pmatrix} 1 & 7 & 17 & 3 \\ 0 & -12 & -30 & -3 \\ 0 & 0 & 6\lambda & -15\lambda \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

При $6\lambda=0$ и $-15\lambda=0 \Rightarrow \lambda=0 \Rightarrow$ III строка станет нулевой строкой $\Rightarrow$
 $\Rightarrow r(A)=2$

Ответ: при $\lambda=0$ $r(A)=2$
 при $\lambda \neq 0$ $r(A)=3$

1.3.31

$$A = \begin{pmatrix} \lambda & 1 & 1 & 1 \\ 1 & \lambda & 1 & \lambda \\ 1 & 1 & \lambda & \lambda^2 \end{pmatrix} \sim \begin{pmatrix} 1 & 1 & 1 & \lambda^2 \\ 1 & \lambda & 1 & \lambda \\ 1 & 1 & 1 & 1 \end{pmatrix} \begin{matrix} \\ \text{II}-\text{I} \\ \text{III}-\lambda\text{I} \end{matrix} \sim$$

$$\sim \begin{pmatrix} 1 & 1 & \lambda & \lambda^2 \\ 0 & \lambda-1 & 1-\lambda & \lambda-\lambda^2 \\ 0 & 1-\lambda & 1-\lambda^2 & 1-\lambda^3 \end{pmatrix} \xrightarrow{\text{III}+\text{II}} \sim \begin{pmatrix} 1 & 1 & \lambda & \lambda^2 \\ 0 & \lambda-1 & 1-\lambda & \lambda-\lambda^2 \\ 0 & 0 & -\lambda^2-\lambda+2 & (1-\lambda)(1+\lambda)^2 \end{pmatrix}$$

$$1) \begin{cases} -\lambda^2 - \lambda + 2 = 0 \\ (1-\lambda)(1+\lambda)^2 = 0 \end{cases} \Rightarrow \begin{cases} \begin{bmatrix} \lambda = -2 \\ \lambda = 1 \end{bmatrix} \\ \begin{bmatrix} \lambda = 1 \\ \lambda = -1 \end{bmatrix} \end{cases} \Rightarrow \lambda = 1$$

$$2) \begin{cases} \lambda - 1 = 0 \\ 1 - \lambda = 0 \\ \lambda - \lambda^2 = 0 \end{cases} \Rightarrow \begin{cases} \lambda = 1 \\ \lambda = 1 \\ \begin{bmatrix} \lambda = 1 \\ \lambda = 0 \end{bmatrix} \end{cases} \Rightarrow \lambda = 1$$

При $\lambda = 1 \Rightarrow \text{II}$ и III строки станут нулевыми строками $\Rightarrow$
 $\Rightarrow r(A) = 1$.

Ответ: при $\lambda = 1$ $r(A) = 1$
 при $\lambda \neq 1$ $r(A) = 3$